

A Packing of 18 Four-Spaces in $\mathbb{R}^{10}$

Limin Ge*

October 9, 2026

Abstract

We give 18 4-dimensional subspaces of $\mathbb{R}^{10}$ whose smallest squared chordal distance is at least 2.540834. This cell of the table of best Grassmannian packings has no published value. The best value implied by known packings is 2.532951, obtained by deleting 1 of the 19 four-spaces of the packing found by this search. The value is checked with exact rational arithmetic from integer coordinates. We do not claim it is optimal.

1 Result

Theorem. *There are 18 subspaces of dimension 4 in $\mathbb{R}^{10}$ whose pairwise squared chordal distances $D(a, b) = 4 - \text{tr}(P_a P_b)$ are all at least*

$$D_{\min} = 2.540834 \dots$$

The exact value of $D_{\min}$, a rational number, is given in Appendix A.

2 Context

N. J. A. Sloane's *Table of Best Grassmannian Packings* (last modified 1997), now maintained by Henry Cohn, lists values for some N in the row $m = 10$, $n = 4$. The cell $N = 18$ is blank in Sloane's table, in Sloane's coordinate files and in Cohn's table as checked on October 8, 2026. The best value implied by known packings is 2.532951, obtained by deleting 1 of the 19 four-spaces of the packing found by this search. The Rankin simplex bound for this cell is $4(10 - 4)/10 \cdot 18/17 = 2.541176$, and the value above lies below it.

3 Verification

Each subspace is given by an integer $m \times n$ basis Y (coordinates times 10^{12} , rounded). With $G = Y^\top Y$ the projection is $P = YG^{-1}Y^\top$, so for two subspaces

$$\text{tr}(P_a P_b) = \frac{\text{tr}(\text{adj}(G_a) A \text{adj}(G_b) A^\top)}{\det G_a \det G_b}, \quad A = Y_a^\top Y_b,$$

an exact rational number. No orthonormality is assumed and no floating point is used. Run

`python grassmann/verify.py grassmann/packings/m10n4N18.txt`

in <https://github.com/Ge-limin/math>.

*Found and checked by Claude, a model released by Anthropic, running on the author's laptop. The author sent the prompts.

4 How it was found

Many random starts of gradient descent on the Grassmannian against a log-sum-exp of the pairwise overlaps, sharpened over time, then a polishing stage (`grassmann/solve.py`); about 65 seconds on one laptop, with about ten such searches running at once.

A Exact value

numerator:

```
158802184928489146239159665217755527389525785880890376815869787741783727
788902168632436533935339974981069894417090539660938052911121216431158997
500388867389350425906752542846468209209577646823
```

denominator:

```
625000000001855251979654227144534174540439435481835787116133774318409443
143198582193254493844913828186759040516122689001466534547010859237587617
90121017623651166607561530750530000150346566712
```

B Coordinates

18 blocks of 10 rows with 4 integers each; divide by 10^{12} . File: `grassmann/packings/m10n4N18.txt`.

```
-85593453256 388503353959 -164274609262 49064505709200982623422 378752265084 -105411461437 -1212662301285130468652 433495958870 493363647737 2087920172
316474022240 -316973534603 -179647091718 5948125806292312732366 -183689909152 -262564242907 59126858583279735956 -124078734944 343627716844 606354556474
-132602858899 415269401666 272227574178 -2140578692955133185911 -272499423111 301185517959 -205544476076523952964 -206402405737 638088210343 -228832248525
145334353093 514139076858 179789359121 50125449143 116042220863 -23600142661 130747335430 -2100219088900185769777 433227578955 224911914866 38171164452
-262902511727 -297607399736 -294091582906 -119113685872734478591 -97564597272 385413106048 -65909033962599203451 193902753080 -158479175243 -303644611659
408230465022 359660279554 -263235269582 17018925331229979239424 -596496904673 125208050677 29446405611-222323588282 209053831730 163534584491 -564374838339
338095142508 26713605732 210321877110 -2297698174420112485637 133031207570 422934185799 -391204413241-119982983468 516459848939 -245741987373 104496803455
642645522216 -213843016264 119690768339 -2888446487978438810973 432758525126 313944976622 3932437446688735672815 -13003828025 114363454635 -243493242997
298578420245 28888676126 -114003861895 -8645872993 406268757438 414927314695 -114267224507 -11690218245569152496 395291595221 102919277219 310975133359
-39898029243 -215876277069 779081504441 4215204460253708547081 74313585412 597853468222 471995212211 -29447344921 248878545364 -211779740845 29588400561
158819386050 463939835804 -635950830380 40081593973-73336397403 219854608342 566618938035 59550419415 -462050947960 46605040263 31054111908 241932226897
282848295022 313813012972 149129932909 15924160665107558850626 -130559382862 26212638931 24683567610426487429507 -129059387530 115955974966 597154763689
98213899091 361187038534 17033238976 -319180000618 64822949825 226512780982 -608129101032 462055442811-44945139981 -457611225351 451674962768 104096644170
-196347507959 481492387833 472507402434 -32291365070440256183 -66854507152 -74645450400 290927418412 333753886718 348391757890 -135354708337 -274197537395
-784490795910 211076210033 -1072925785 -4329470952538690787635 -70371990239 276479186382 -1422193082005050656508 -108652100658 -245251473425 195896166695
-397241605901 -321280393816 -163589263679 54026452394765046550 -678237798906 19893264993 253588154197232276666321 -451944115816 -229222359969 -171872654335
112825466379 -238333935772 70303887476 -6235630177054278803367 216506981914 -288310967642 -606082453270092514249 336031660049 248275204147 -337331277885
53460460164 -229241023878 -164708167820 -149918078150287607761 -46431541516 -282657986742 -24003461220033856640197 -77465076823 -767341766052 83846764869
172597274642 -95474003396 537378308992 28950938387530021449357 552943863888 212584533836 32451766221752045642168 -409626615334 9495701276 -166638198564
167244743278 8909711423 -46518441367 -519481382650 17163846476 -243112786402 117932142231 -150314169828785965873 384208215306 -4835095825 531364382771
186856447221 -600030547629 225584114855 1862016606283665306214 -63472227556 -216992933455 6302586783303899316158 203453555535 -67798863389 -145613692612
-165912747500 -281286703326 73683877368 -530351151724807019725 -430491639448 252185143060 14043087613086652028470 -32746691775 -424486055246 511080526782
-40462120788 -405717771274 170687997674 42475193250822512643582 379231631653 -93614994509 96233478944-152938626455 -443260303500 -475931704756 -141005430350
-82845669072 -22863926756 -300191052001 5489369294463251926192 -610526096545 -123962816801 -36679946105276056860 -486570096283 431240749166 -227429307430
659736085950 -98838185711 -542059876866 1044306320 2966279596314 47935176608 -534944595966 -4541450286653529749839 -418514883352 67352453853 119687450686
341150970688 -193104170718 -255295495087 -12970000435262646226 -193913298894 329082649740 11927931887407105884855 460216991366 -369694344608 202480799878
-66337633449 518209937812 100217770095 380690009700-245213516264 161820542795 -438336545359 30677602209226560656962 507953264173 117464002858 -113438327502
-6944305684 50400146071 666417710401 445708207102 -55980045182 101035910850 500687347428 -3088840778321970285945 265154028379 137030907955 759477121140
431777060367 -445232541593 2746656501554 -10204705375749988235181 316265075912 67444630051 17221712504 -416214876370 -299423941688 -186701858014 -40824205924
400496159893 225996074880 -9780050424 93479582450 252521028742 48667185515 -48072168880 562976210564 455400840413 -156367679449 -771997532942 18035494292
-266119085058 -220244182065 266526747617 -18326951670425715489 227535938583 -341603515278 -5231746779601334358835 -330360919247 156126766817 -101396568430
-24734162401 118001957769 -146633648119 -94267105537665330691 -609805380025 323886124147 39761790927-101590165526 58920898774 -203944817400 -38945457826
-515828753518 -542510624487 -95763796615 360069222306767909835 577786470849 534059421739 108309100674-223595244954 1216134478 -174225138650 137566752680
-302341560636 229384721424 377395845737 -668113142282089094056 191910555645 75853860870 -105506177935-149092702351 -479966746065 113720978064 563332348390
207006129302 -181537140190 401247992590 -44115376869207973325197 -17119960791 -15819992675 -425273539208613008015 52839662063 301093410082 -138816138094
648737389659 34644831476 375327324483 -28610547394504495590651 -133478023093 82360216279 420769750159-438111110794 275635111688 621010017856 -304756929242
278564123857 364920978393 128064259441 233363277197-101252611114 124609156568 -122271682029 -538741803382664703057 -11038683397 154376092883 654538645135
-187524695779 158415362202 -10529758278 5292432486914008906109 -128923424734 238192859836 -220144348709568730525 307694767240 189746986384 139024914610
496628250137 -133452975889 -440128126404 -619031206633927586898 -301569168849 -812039948049 2266093586677226539481 49112404353 -321178990177 -1556077449316
157342887044 -376509228545 381700533695 26080188043212966752357 641828538413 -130177479990 2090675731224507860786 -471881312478 453241280480 -81674166028
-349154751297 -263708134467 -133174020847 -10635623030741430141 -59666516822 -131591537857 8095520668 201868088891 103411502267 -9725030634 353818301069
96789656408 -39757491580 -74117788430 881711484669 -221407888574 -302379407494 210602439447 -409894549480214491462 2651929226 319472024809 394177793550
37291716113 685468658477 -259976709040 13091264590 -30933666085 -495133810719 423491195475 12383874101127381962440 -306134482226 331325989039 -249621034519
-101743352700 -23793793070 432230845541 233363277197-101252611114 124609156568 -122271682029 -538741803382664703057 -11038683397 154376092883 654538645135
-227958084114 373379200791 474941938131 11659254702-245651598129 294945790400 38732625388 257037273276-159477801900 593575118512 183774248148 25932053701
-2203177828428 -37001391051 -91414788903 49591917288 49406906937 -274319783555 -158880752105 -81523630490536051096002 20692520362 -107680349940 -359070047967
-421381502607 80988761366 -95314315990 20193379684043861285206 192375538737 520251488204 492509721757-72957951667 20605663781 754408161750 342350982354
188402079383 -428369265464 437726513600 30952119388063144646202 -179899919929 -317484668075 3627032297723723399086 -400692819900 -247447095941 518680389158
-55193122456 291085288096 212610613931 473328415055-209056057832 -495133810719 239272203021 -259380564773656437294 629538202477 111407807856 -120163146240
325475865920 403407310194 507084278100 190507375036-345003526900 543856058822 -192500466966 2429192237209550550107 -381001631762 126754681855 145964521529
158443390899 -20394679363 -81552381458 5117480220249232552405 369900451754 79161105638 -323367589347-432580638708 -45203767619 133592617382 -505334165746
-660464380696 56764058321 99058823882 352601806976 6988827244861 -118357805327 -492730345557 -1171561913704037072700 -43910784122 215220767382 5387674232
56726662137 -407414973903 29376263261 -129924069741104601379175 19010893766 34292958473 -28976097771-1281163692653 56755465778 -347059040410 198856318650
108605536758 265149482000 259248046422 -777334652853715091624 -194961858849 -166825972747 469717576515051953916 228221023877 51645668656 362302502403
389858046670 264850429593 -635673125631 4375960896480284631130 349224992893 -338407175974 -28938413932819875113 67446915187 -377208406514 151257041087
```